Integrate Engineering Design

👨‍🎨 👩‍🎨

DAEN 640 Capstone Senior Design

Alexander Abuabara

Jan 2026

Plan for today

Engineering Design Process

Integrated Engineering Design Process

System Architecture, Data Pipeline

The Toaster Project

Challenges, Solutions

Learning Goals

Map the V-Model

Identify Configuration Items

Understand Traceability

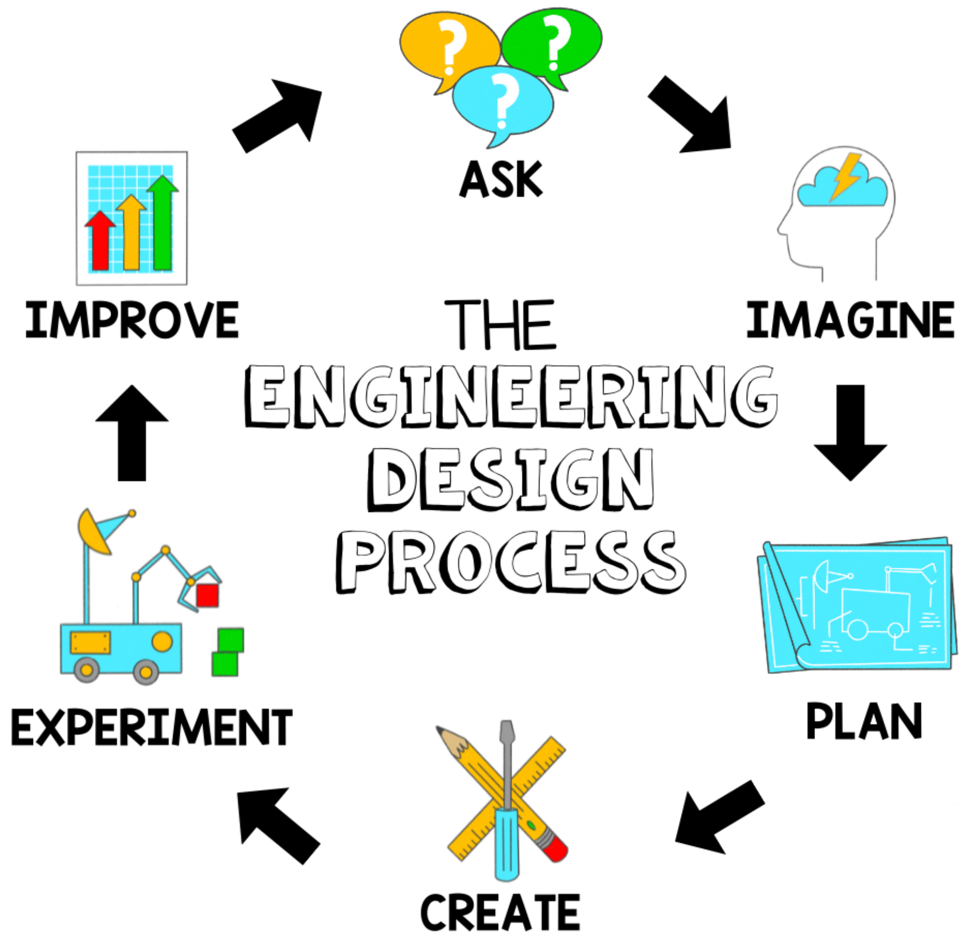
Connect Tests to Artifacts

Evaluate Change Impact

Descriptions

- **Map the V-Model**
  See how requirements, design, and implementation connect to testing and validation.
- **Identify Configuration Items**
  Recognize key items (requirements, schemas, code, pipelines, infrastructure) and how they are versioned.
- **Understand Traceability**
  Follow how requirements link to system components, data pipelines, and tests.
- **Connect Tests to Artifacts**
  Identify what is tested or validated for each major configuration item.
- **Evaluate Change Impact**
  Understand how changes ripple across design, implementation, and testing.

Engineering Design Process



A step-by-step approach engineers use to identify problems, brainstorm solutions, build and test prototypes, and improve designs to create effective, practical solutions.

https://www.youtube.com/watch?v=qqs3fxfmWr4

Life Lessons from a Pencil

What matters is what’s inside

Keep yourself sharp

Mistakes are part of the process

Pressure creates output

Collaboration matters

Durability comes from design

Imagine a pencil journey as a metaphor

Raw Materials → Shaping → Testing → Final Pencil

Engineering Design Process

1. Define the problem (*Why make a pencil?*)
2. Research & brainstorm (*What materials & designs?*)
3. Develop concepts (*Sketch ideas, pick the best*)
4. Build & prototype (*Make a sample pencil*)
5. Test & evaluate (*Does it write? Break?*)
6. Redesign/refine (*Improve core, casing, process*)

Engineering Design Process

Material Selection and Functionality

Mapping Process to Design Stages

Balancing Constraints

Engineering Design Process

Material Selection and Functionality

Multiple cycles of design thinking. Define the problem, explore materials and methods, prototype, test, and refine.

Mapping Process to Design Stages

Engineering design is not linear. Feedback from testing and production continually informs improvements.

Balancing Constraints

Deliberate choice that balances functional requirements, "manufacturability", cost, and sustainability. How engineers translate design goals into real-world material and process decisions.


THE ENGINEERING DESIGN PROCESS
DEFINE
the problem
IDENTIFY
constraints on your solution (e.g. time, money, materials) and criteria for success
BRAINSTORM
multiple solutions for the problem
SELECT
the most promising solution
PROTOTYPE
your solution
TEST
and evaluate your prototype
ITERATE
to improve your prototype
COMMUNICATE
your solution

Integrated
Engineering Design Process

Integrated Engineering Design Process

(multiple disciplines and feedback loops)

Materials + Mechanical + Environmental + Economics

📝💡🔧🔬♻️💰

Integrating engineering design means:
Combining different subjects, ideas, and people with the Engineering Design Process to solve problems together

Instead of working in separate steps, this approach encourages teamwork and big-picture thinking to create complete solutions

It helps us build skills in STEM, critical thinking, and practical problem-solving, from the first idea all the way through testing and improving designs

You see this approach in schools and in big projects
(like designing buildings :)

System Architecture, Data Pipeline

System Architecture (V-Model Aligned)

Objective: Provide a modular, testable architecture that supports traceability from requirements through validation

High-Level Architecture

- Presentation Layer: Dashboards, APIs, reporting interfaces
- Application / Processing Layer: Business logic, analytics services, orchestration
- Data Layer: Raw, curated, and analytics-ready data stores
- Infrastructure Layer: Compute, storage, networking, security controls

V-Model Mapping

Decomposition & Design *(Left Side)*

System Requirements → System Architecture
Subsystem Design → Component Design

Integration & Verification *(Right Side)*

Component Testing → Subsystem Integration
System Verification → User Acceptance Validation

Configuration Items

Architecture design documents

Interface control documents

Infrastructure-as-Code templates

Versioned service definitions and APIs

End-to-End Data Pipeline

Data Ingestion

- Batch and streaming sources
- Schema validation and data quality checks

Data Processing

- Transformation, aggregation
- Business rule enforcement

Data Storage

- Raw (immutable), curated, and analytics layers

Data Consumption

- BI tools, ML models, downstream services

V-Model Alignment

Requirements → Data definitions, quality rules

Design → Pipeline workflows, storage schemas

Implementation → ETL/ELT jobs, orchestration logic

Verification & Validation:

Data quality tests, Pipeline integration tests, End-user data validation

Key Configuration Items

Data schemas and metadata definitions
Pipeline code (ETL/ELT, orchestration)
Data quality rules and test cases
Versioned datasets and lineage artifacts

Outcome

Full traceability from data requirements to validated outputs
Controlled change management and audit readiness

Project Alignment Questions

1. *V-Model Mapping:* How does your project diagram map requirements → design → implementation (left side) to testing → integration → validation (right side)?
2. *Configuration Items:* What are the key configuration items (e.g., requirements, schemas, code, pipelines, infrastructure), and how are they identified and versioned?
3. *Traceability:* How does the diagram show traceability between requirements, system components, data pipelines, and verification activities?
4. *Verification & Validation:* For each major CI, what test or validation activity is defined and where is it represented on the right side of the V-Model?
5. *Change Impact:* How does the diagram communicate the impact of changes to requirements or data definitions across design, implementation, and testing?

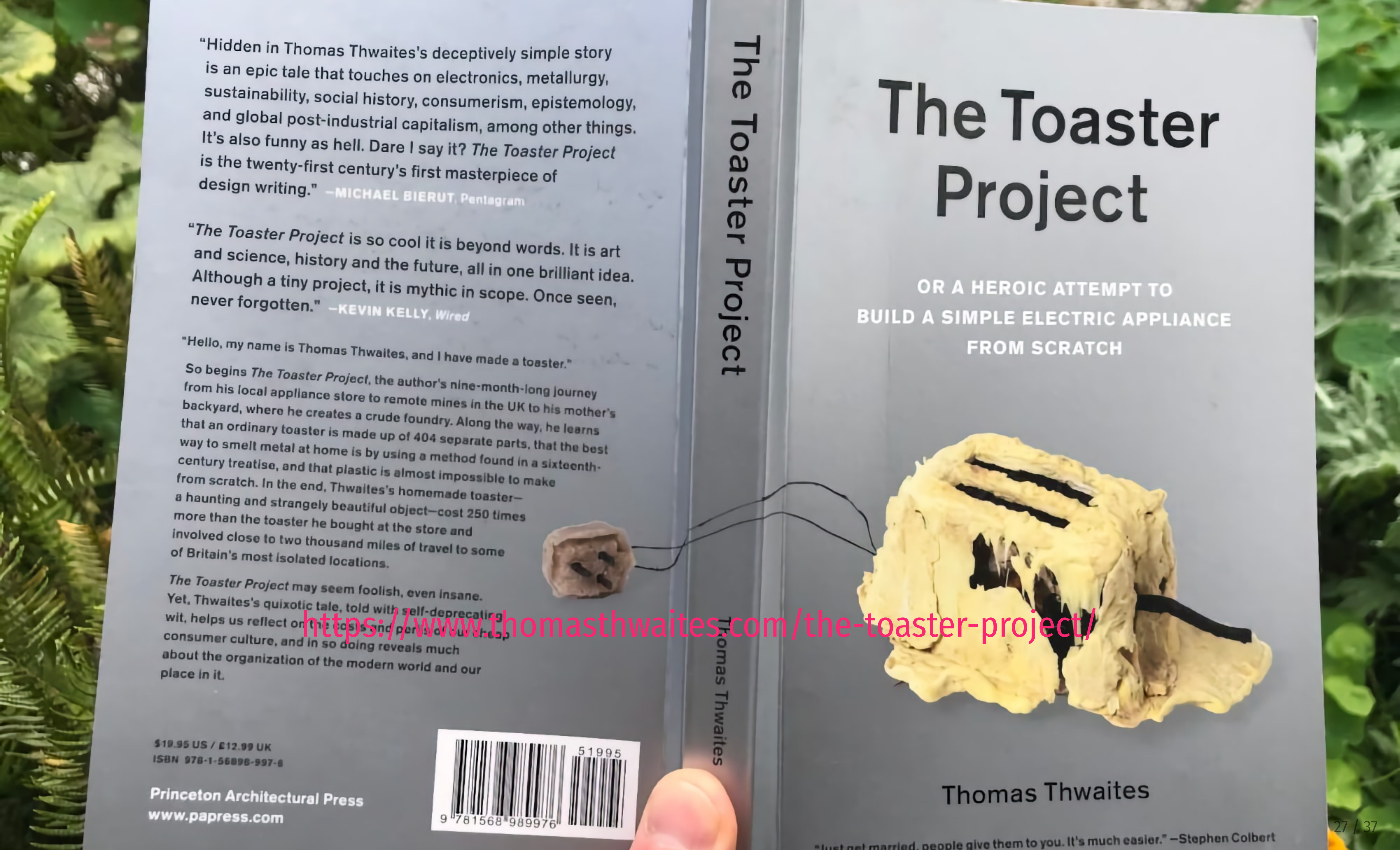
Mini Project Packet

Please be sure to include these answers in your submission

On Canvas - Due 2/16 at 5 PM

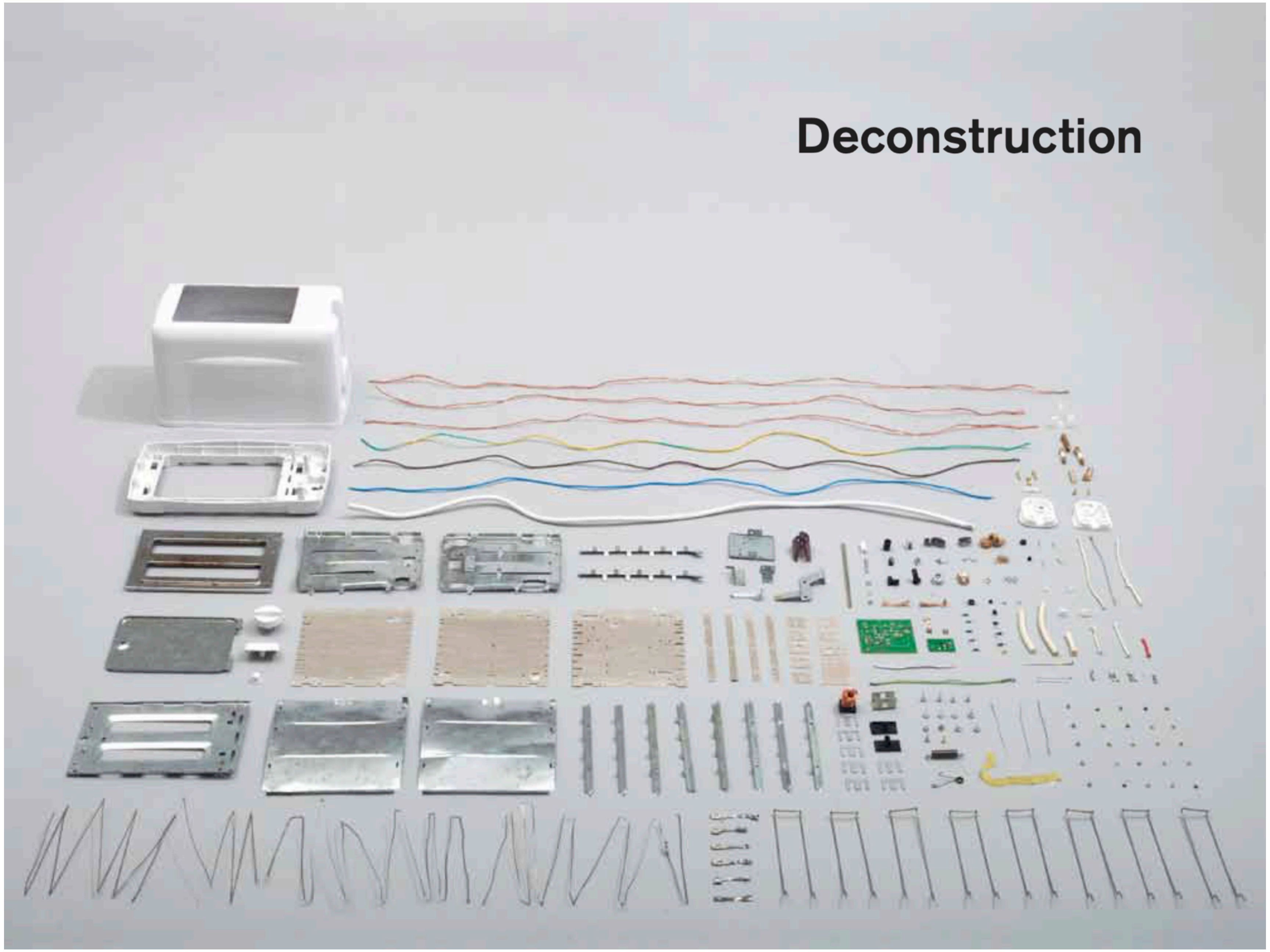
https://canvas.tamu.edu/courses/431435/assignments/2873361

The Toaster Project



https://www.thomasthwaites.com/the-toaster-project/

Deconstruction



My toaster


Toasters
£1187.54
£16.45
£24.45
£22.95
Breville
Breville
Breville
2 slice brushed stainless steel toaster

Challenges, Solutions

Q1. Feasibility & Hidden Complexity

What assumptions are we making about the simplicity, availability of resources, and manufacturability of our design, and how will we validate or challenge them early?

Expose hidden complexity and prevent unrealistic scope definition.

Q2. Materials, Supply Chains, and Constraints

What critical materials, tools, and external resources does our project depend on, and what are the risks if they are unavailable or delayed?

Early supply-chain awareness and contingency planning.

Q3. Cost and Schedule Realism

How will we estimate cost and schedule realistically, and where do we expect uncertainty, rework, or iteration to occur?

Encourage defensible planning with buffers and explicit risk acknowledgment.

Q4. Performance Definition & Validation

What does “success” mean for our system, and how will we verify and validate performance throughout the project rather than only at the end?

Link requirements to measurable, testable outcomes.

Q5. Process Choices & Tradeoffs

How will we evaluate alternative technical approaches, and what criteria will justify simplifying, changing, or abandoning a design when constraints arise?

Promote pragmatic, evidence-based engineering decisions.

Q6. Documentation & Learning from Failure

How will we document decisions, failures, and lessons learned in real time, and how will this knowledge be used to improve subsequent design choices?

Reinforce engineering as an iterative, learning-driven process.